6 **(a)** State what is meant by a *photon*.

.....

..... [1]

(b) An X-ray photon of wavelength 965.0 x 10^{-12} m collides elastically with a stationary electron, as shown in Fig 6.1.

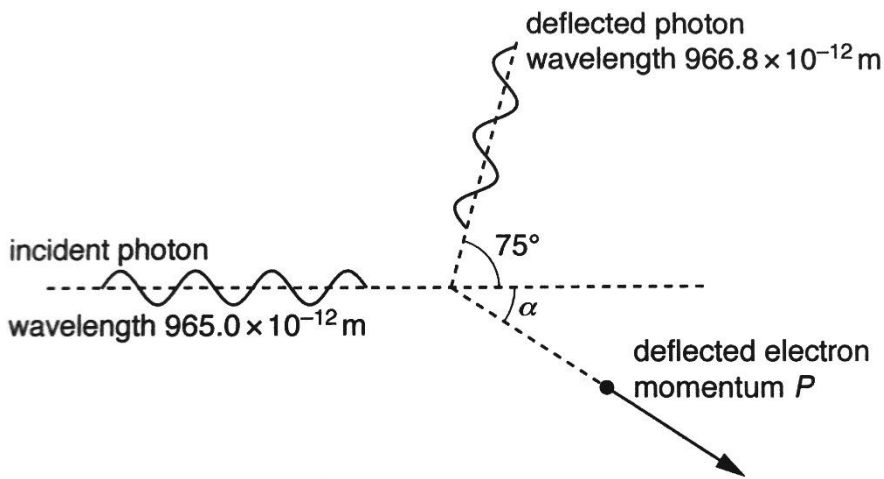

Fig. 6.1

The photon is deflected through an angle of 75° and its wavelength changes to 966.8 x 10^{-12} m after the collision. The electron is deflected through an angle α.

(i) Calculate the change in the energy of the photon as it is deflected.

energy change = …… J [2]

(ii) Use conservation of energy to show that the momentum *P* of the deflected electron is 8.36×10^{-25} N s.

[2]

(c) Momentum is a vector quantity. By taking components of momentum perpendicular to the direction of the incident photon, calculate the angle α of deflection of the electron as shown in Fig 6.1.

Explain your working.

angle α = …… ° [5]